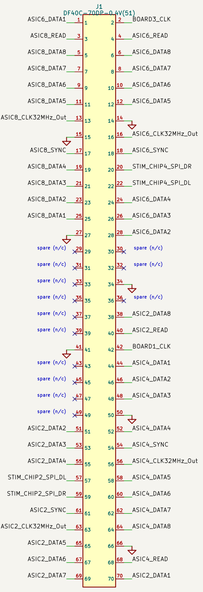
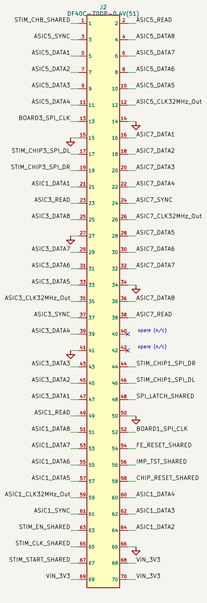
FPGA–LDO PCB V33 VERSION B – LDO BOARD (REMOVABLE)

FRONTEND MATING (TOP FACE)

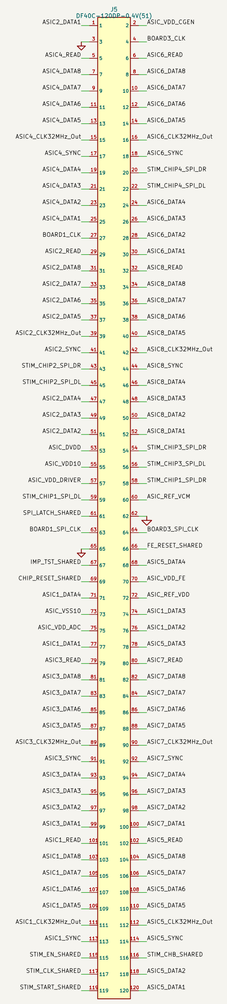




1. Function: Mate the Artix Frontend BL V1 bottom-side receptacles J1/J2 (DF40C-70DS) with two DF40C-70DP plugs on this board's top face. 1.5 mm stack, pin n to pin n. J1 runs along the left long edge and J2 along the right long edge, parallel. pin 1 at the ASIC end. Carries the 108 ASIC signals the two PCB-5 pairs use (J1 50, J2 58), VIN_3V3 (J2.68–70, three 0.3 A contacts) and 14 GND contacts.

2. Source / changes: Claude design. pin map V5 (10 Nov 2026), from the layout V11 lane plan (plan_v11.py: 2026-11-10_FPGA_Work); Howard, 10 Nov 2026: one Frontend connector on each long edge, and "B: both on J2" – every ASIC group sits whole on one connector. J2 carries ASIC1, ASIC3, ASIC5, ASIC7 and the 14 slow nets on the east half of the ports (8 shared controls, STIM_CHIP1/3 SPI, BOARD1/3_SPI_CLK); J1 carries ASIC2, ASIC4, ASIC6, ASIC8, BOARD1/3_CLK and STIM_CHIP2/4 SPI. Rows nearest the ASIC end go to J3 (L1/L2), the far rows to J4 (L5). Changed from V29 (map V4): ASIC1 and ASIC5 READ and DATA5–8 move from J1 to J2, and most contacts are re-ordered. Unchanged: VIN_3V3 on J2.68–70, GND 14/15/27/34/41/50/66 on both. Unlisted pins (13 on J1, 2 on J2) are no-connect. The Frontend has NOT adopted this map yet; it needs its 25 MHz clock input moved to another bank (Frontend chat, 10 Nov). Hirose DF40 catalog, June 2024 (LCSC copy). URL https://datasheet.lcsc.com/datasheet/pdf/145ab5d527b34eafa7a47375cb561c6a.pdf?productCode=C5640153 . DF40C-70DP C2692097 / 70DS C294854 (JLC, read 2 Nov 2026). Unverified: two rigid DF40 pairs between two boards (the DF40 drawing, sheet 3 item 4, advises one pair per board pair).

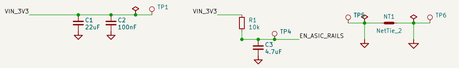
ROUTER LINK – J5 DF40C-120



1. Function: J5 on this board's top face, at its north end, takes the router from above. pin n to pin n, 1.5 mm mated. It carries the 100 port signals, the 8 shared controls, the nine ASIC rails, 1 AGND and 2 GND: exactly the version A port nets except AC_IN, whose default (R6/TP7) sits on the router. So B = A split at J5.

2. Source / changes: Claude design (Howard chose one DF40C–120 on 7 Nov 2026). Hirose drawing EDC3-311351-00 DF40C-*DP-0.4V(51). rev 2 (2024-07-08. HRS Drawing for Reference. printed 1 Oct 2026). note 2: the four corner metal fittings have no electrical continuity: they are separate pads 0.475 mm beyond the contact rows (KiCad MP pads), so every numbered contact is live. DF40C-120DP 25.52 x 2.97 mm (same drawing, table): 0.3 A per contact and 30 mating cycles (Hirose DF40C-120DP product page CL0684-4107-9-51). Stock KiCad 10 footprints (120 contacts + 4 MP fittings on the header). Pin order: J5 pin map V4 (10 Nov 2026, version B floorplan V1): column 1 west, odd pins = north row: J1 nets on columns 2–26, J2 nets on 27–60, rails straight under their J3/J4 pins; lanes checked on signals-only boards, not a layout. Unverified: signal return with only 2 GND + 1 AGND contacts for about 100 signals; rail current per contact (a few mA per rail).

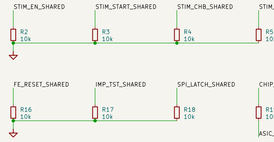
VIN_3V3 ENTRY, DELAYED ENABLE, RETURN BOND



1. Function: C1/C2 hold VIN_3V3 for the converter and regulators; it arrives on J2.68–70, the three Frontend supply contacts. R1/C3 delay EN_ASIC_RAILS by about 20 ms after VIN_3V3. so all nine ASIC rails start after the frontend 1.5 V bank supply. NT1 is the only AGND–GND bond for all four ASIC boards (the ASIC board must not add its own).

2. Source / changes: Claude design, no manufacturer reference circuit. Delay: tau = R1 x C3 = 47 ms; LT3094 EN/UV turns on at 1.26 V, so t = 47 ms x ln(3.3/2.04) = 23 ms; LT3042 threshold assumed similar (to confirm). Nine EN/UV pin currents x 10k must stay well below 1.26 V. The frontend VCCO_1V5 start time is not measured. NT1 follows the LDO V13 NT100 single-bond idea: PCB–5 J3/J4 carry AGND (42/44) and DGND (46/48) separately and no PCB–5 part joins them; in version B the router carries both returns through J5 and adds no bond (checked in its PCB file). Layout: C901/C1001 return to AGND and the LT3582 output caps to GND, so place NT1 beside U9/U10 to keep converter ripple current out of the ASIC AGND. Unverified: bond location in layout, and whether VIN_3V3 sags at the J1 contacts (0.9 A for three).

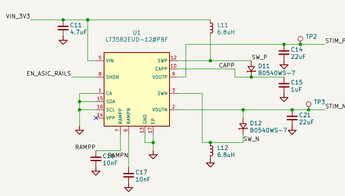
ASIC CONTROL SAFE IDLE



1. Function: Hold every FPGA-to-ASIC control line at its safe level while the FPGA is unconfigured: STIM_EN/START/CHB/CLK, FE_RESET, IMP_TEST and SPI_LATCH low; CHIP_RESET high (to ASIC_DVDD), so the ASICs stay in reset until the FPGA releases them. About 150 uA per line when driven to the other level.

2. Source / changes: Claude design from Gerald's firmware (Archive/.../07_FPGA_512_Reference/repository/fpga): top_custom.v section 3b drives CHIP_RESET high for 2^20 cycles (~65 ms) at boot then low (active high); line 264 holds FE_RESET low ("amplifier reset released; normal operation"); STIM_* held low; custom_board.pdf line 100 leaves IMP_TEST an input with PULLMODE=DOWN; spi_programmer.v keeps SCLK idle low and pulses LATCH high. Needed because the Frontend ties PUDC_B high (R25 470R), disabling FPGA pull-ups during configuration (AMD UG470. PUDC_B). V23 adds R16–R23 to V22's R2–R5: V28 (Howard, 9 Nov 2026, "Drop R20–R23") removes the four board SPI/recording-clock pulls; while the FPGA configures, CHIP_RESET (pulled high) holds the ASICs in reset and SPI_LATCH (pulled low) blocks any SPI update, so floating clocks have no effect (Claude judgment, not a Gerald rule): it also frees bottom space at the bus taps. READ/SYNC are ASIC outputs and SPI data is only latched by SPI_LATCH, so they get no pull. Unverified: ASIC input leakage and thresholds: reset polarity is from firmware, not an ASIC data sheet.

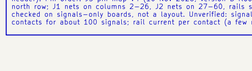
STIMULATION SUPPLY – VIN_3V3 TO +/–12 V (LT3582-12) THEN +/–10.0 V LDOs



1. Function: LT3582-12 makes STIM_POS_12V (+12 V. boost through D11 to CAPP and its output-disconnect switch to VOUTP) and STIM_NEG_12V (–12 V. single-inductor inverter through D12) from VIN_3V3. as pre-regulators for the stimulation LDOs. Fixed +/–12 V with internal feedback and compensation; starts with EN_ASIC_RAILS (SHDN on above 1.1 V). About 10 mA per output.

2. Source / changes: ADI LT3582/LT3582-5/LT3582-12 data sheet 35825212fb. Figure 13 p.22 (+/–12 V from a single 5 V input). Pin Functions p.8, Choosing Inductors p.18, Capacitor and Diode Selection p.19. https://www.analog.com/media/en/technical-documentation/data-sheets/35825212fb.pdf . Changed from Figure 13: input 3.3 V instead of 4.5–5.5 V (allowed 2.55–5.5 V; max load at 3.3 V about 40 mA per output from the p.19 equations, need about 10 mA); C1 –> C11 4.7 uF at 10 V 0603; C2 –> C21 22 uF at 25 V 0805 and C3 –> C14 22 uF at 25 V 0805 (about 6–7 uF effective at 12 V. inside the 2.2–10 uF range); C4 –> C15 1 uF 25 V on CAPP; C5/C6 –> C16/C17 10 nF; L1/L2 6.8 uH (Coilcraft XFL2010-682 in the figure) –> Murata DFE201610E-6R8M (1sat to confirm >= 0.42/0.69 A peak per p.18); D1/D2 B0540WS-7 as in the figure; SHDN to EN_ASIC_RAILS instead of an input logic signal; CA/SDA/SCL to GND, VPP open (I2C/OTP unused on the –12 version). V24 replaces the TPS65130 (Howard. 8 Nov 2026: smaller parts) and removes R11–R15, C12. C13, C18–C20. C22. Unverified: JLC stock, ripple after layout, start-up current at the J1 contacts.

STIM SUPPLY SELECT



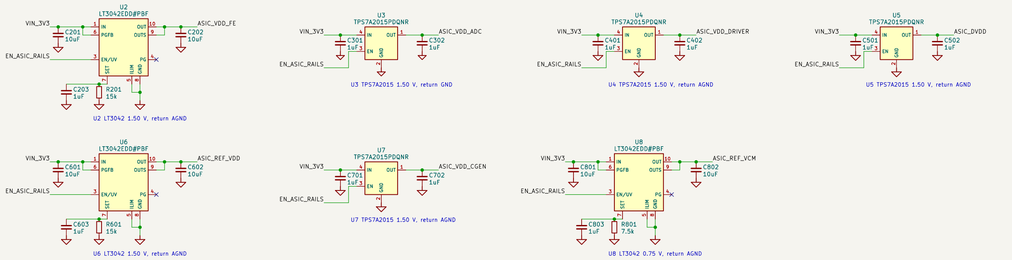
1. Function: JP1/JP2 connect the on-board LT3582-12 outputs to the stimulation LDO inputs. Cut both to feed STIM_PRE_POS/NEG from a bench supply on TP8/TP9 (return: TP6 AGND), as LDO Board A did.

2. Source / changes: Claude design. LDO Board A took INPUT_VDD10/INPUT_VSS10 from outside (Reference/LDO Board A J15/J8). Limits: LT3042 IN <= 20 V, LT3094 IN >= –20 V (ADI data sheets, Absolute Maximum Ratings); keep >= 1.5 V headroom over +/–10.0 V. Leave U1 unfitted, or its outputs open, when external +/–V is used. Unverified: external source noise.

1. Function: U9 (LT3042) makes ASIC_VDD10 = +10.0 V from STIM_PRE_POS; U10 (LT3094) makes ASIC_VSS10 = –10.0 V from STIM_PRE_NEG. Both are referenced to AGND and start with EN_ASIC_RAILS. Each serves the stimulators of all eight ASICs (about 8 mA at Gerald's 2 mA per board maximum).

2. Source / changes: U9 inherits Gerald LDO Board A U8 (LT3042. RSET 90k. input from an external INPUT_VDD10). Changed: input is now STIM_PRE_POS (on–board +12 V through JP1. or external): RSET 100k (E96: LDO A 90k). U10 replaces LDO A U9. which floated a positive LT3042 on an external INPUT_VSS10. with a true negative LDO: ADI LT3094 data sheet Rev 0 (2018). Typical Application p.1 and Pin Functions p.15. https://www.analog.com/media/en/technical-documentation/data-sheets/lt3094.pdf . Changed from p.1: RSET 100k (was 33.2k). CSET 1 uF (was 4.7 uF); PGFB to IN. ILIM to GND. PG and VIOC open. per Pin Functions: EN/UV driven positive (turns on above +1.26 V). V23 (Howard. 8 Nov 2026): +/–10.0 V per Gerald's Nov 2026 pin sheet. so R901/R1001 = 100k (V19–V22 used 90.9k for +/–9.09 V. after LDO A's 90k). About 2 V headroom below the +/–12 V pre-regulators. Confirm the ASIC board VDD10/VSS10 0201 capacitor rating (>= 16 V). V20 output capacitors (Claude change after the 7 Nov review): ceramic capacitance falls with DC bias. so V19's 10 uF 25 V 0603 likely kept only about 3–4 uF at 9.09 V. LT3042 needs >= 4.7 uF and LT3094 >= 10 uF ceramic output capacitance (ADI data sheets, front-page Features). C902 is now 22 uF 25 V 1206; C1002 22 uF 1206 plus C1004 10 uF 25 V 0805. V22: C901/C1001 10 uF 25 V 0805 and C903/C1003 1 uF 25 V 0603 (the SET pins sit at +/–10 V). Unverified: effective capacitance from Samsung DC-bias curves at 10 V.

ASIC RAILS – THREE LT3042 (ANALOG / REFERENCE) + FOUR TPS7A20 CELLS FROM VIN_3V3



1. Function: Seven regulators make the ASIC rails from VIN_3V3: each rail is shared by all eight ASICs through the two PCB–5 ports (version B: through J5 and the router). LT3042 (U2 VDD_FE. U6 REF_VDD. U8 REF_VCM 0.75 V): Vout = 100 uA x RSET. TPS7A2015 (U3 VDD_ADC. U4 VDD_DRIVER. U5 DVDD. U7 VDD_CGEN): fixed 1.50 V. All start from EN_ASIC_RAILS (TPS7A20 EN high above 0.9 V max, LT3042 EN/UV above about 1.24 V).

2. Source / changes: LT3042 cells inherited from Gerald LDO Board A (Reference/LDO Board A/PCB.kicad_sch): IN. EN/UV and PGFB at the input. ILIM to ground. OUT=OUTS. CIN/COUT 10 uF at 10 V 0603. CSET 1 uF. PG open. RSET 15k/7.5k: ADI LT3042 data sheet. Typical Application p.1 and Pin Functions. https://www.analog.com/media/en/technical-documentation/data-sheets/lt3042.pdf. Claude changes: EN/UV to EN_ASIC_RAILS: one regulator per rail for eight ASICs. V24 (Howard. 8 Nov 2026, "find smaller parts"): U3/U4/U5/U7 changed from LT3042 3x3 DFN cells to TI TPS7A2015PDQNR 1x1 mm X2SON (TI SBVS338H Rev H, Jul 2024: Pin Functions p.4 DQN 1 OUT/2 GND/3 EN/4 IN/5 pad: Recommended Operating Conditions p.5 CIN 1 uF, COUT 1–200 uF with >= 0.47 uF effective: Electrical Characteristics p.7 VEN(HI) 0.9 V max: https://www.ti.com/lit/ds/symlink/tps7a20.pdf): their RSET/CSET (R301–R701. C303–C703) are removed and CIN/COUT are 1 uF 10 V 0402. Trade-off: 7 uVrms against 0.8 uVrms (LT3042), accepted by Gerald 9 Nov 2026 ("I think its okay to do that"). VDD_ADC still returns to GND as on LDO Board A. REF_VCM stays LT3042 because 0.75 V is below the TPS7A20 0.8 V minimum. Loads of REF_VDD/REF_VCM unknown.

Frontend J1/J2 on top: J5 DF40C–120 on top (north end) to the router.
3x LT3042 + 4x TPS7A20 ASIC rails: LT3582–12 or external +/–V: LT3042/LT3094 +/–10.0 V.
Schematic only. no layout. Input: VIN_3V3 from Frontend J2.68–70.

Rice – FPGA PCB workspace

Sheet: /
File: FPGA_LDO_V33B_LDO.kicad_sch

Title: FPGA–LDO V33 B – removable LDO board

Size: User	Date: 2026-11-10	**Rev: V33**
KiCad E.D.A. 10.0.6		Id: 1/1